

BAYEVO, Russian Federation

WMO: 298270

Lat: 53.2781N

Lon: 80.7755E

Elev: 122

StdP: 99.87

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.4	-32.9	-39.1	0.1	-35.9	-35.7	0.1	-32.6	13.3	-8.5	11.6	-8.3	1.5	250	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.4	18.7	28.7	18.3	27.2	17.6	20.6	26.8	19.7	25.9	18.9	25.0	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.7	23.1	17.5	12.7	22.3	16.6	12.0	21.6	60.1	26.9	56.9	25.9	54.3	25.0	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.0	9.4	8.2	DB	-39.9	33.3	4.5	1.8	-43.2	34.5	-45.9	35.6	-48.4	36.6	-51.7	37.8
(5)				WB	-39.7	22.5	3.9	1.8	-42.5	23.8	-44.8	24.9	-47.0	25.9	-49.9	27.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-17.9	-15.3	-7.1	5.0	13.0	18.6	19.9	17.4	11.2	3.9	-5.9	-13.6	(6)
(7)		DBStd	14.67	8.65	7.76	6.94	5.66	5.09	3.81	3.13	3.49	4.31	4.92	7.83	8.24	(7)
(8)		HDD10.0	3747	866	708	532	169	31	1	0	1	36	196	476	732	(8)
(9)		HDD18.3	5936	1124	942	790	401	176	43	20	59	217	448	726	991	(9)
(10)		CDD10.0	1018	0	0	0	18	122	260	308	231	72	7	0	0	(10)
(11)		CDD18.3	165	0	0	0	0	9	52	71	31	3	0	0	0	(11)
(12)		CDH23.3	1672	0	0	0	7	148	474	589	379	74	0	0	0	(12)
(13)		CDH26.7	443	0	0	0	1	32	127	158	104	21	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.4	3.4	3.6	3.8	3.7	3.1	2.7	2.8	3.0	3.4	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	347	16	13	15	21	30	46	65	35	31	28	27	22	(15)
(16)		PrecMax	494	38	34	31	59	89	108	132	90	71	79	56	40	(16)
(17)		PrecMin	180	2	0	1	0	3	9	9	3	6	8	9	8	(17)
(18)		PrecStd	79	8	8	8	12	20	26	33	21	17	18	11	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	1.7	8.2	24.2	30.7	32.0	32.2	32.3	29.4	20.6	10.2	2.4	(19)
(20)			MCWB	0.5	0.4	4.4	13.2	16.9	18.5	20.2	19.2	16.0	11.7	7.1	0.7	(20)
(21)		2%	DB	-0.9	-0.3	5.1	20.3	27.1	29.9	30.1	29.4	25.3	17.1	7.8	0.7	(21)
(22)			MCWB	-1.8	-1.3	2.5	11.5	14.7	18.0	19.6	18.9	14.7	10.0	4.9	-0.4	(22)
(23)		5%	DB	-3.0	-1.8	3.5	17.0	24.9	28.0	28.4	27.3	22.1	14.4	5.2	-1.0	(23)
(24)			MCWB	-4.0	-2.8	1.4	10.0	14.2	17.8	19.1	17.8	13.6	8.8	3.1	-1.9	(24)
(25)		10%	DB	-5.2	-4.0	1.9	14.1	22.3	26.2	26.7	25.3	19.6	11.5	3.0	-2.6	(25)
(26)			MCWB	-6.0	-4.8	0.4	8.5	13.5	17.3	18.6	17.0	12.4	7.4	1.7	-3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	0.6	4.9	14.1	18.8	21.1	22.6	21.5	17.3	12.5	7.8	0.9	(27)
(28)			MCDWB	1.5	1.3	7.7	22.5	27.1	27.5	27.9	27.5	24.9	18.6	9.9	1.9	(28)
(29)		2%	WB	-1.8	-1.2	2.8	12.1	16.5	19.6	21.0	19.7	15.5	10.7	5.0	-0.3	(29)
(30)			MCDWB	-1.0	-0.3	4.7	18.6	24.1	26.2	26.9	27.2	23.4	15.9	7.4	0.5	(30)
(31)		5%	WB	-3.9	-2.8	1.6	10.4	15.2	18.8	20.1	18.7	14.3	9.2	3.3	-1.8	(31)
(32)			MCDWB	-3.1	-1.9	3.2	16.1	22.5	25.4	25.8	25.7	21.2	13.7	5.3	-1.1	(32)
(33)		10%	WB	-6.0	-4.8	0.4	8.8	14.1	18.0	19.4	17.6	13.0	7.7	1.6	-3.5	(33)
(34)			MCDWB	-5.2	-4.0	1.8	13.9	21.0	24.4	24.9	24.1	18.7	11.3	3.0	-2.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	10.0	9.8	10.2	11.7	10.7	10.3	11.8	11.4	9.0	6.6	8.2	(35)
(36)			MCDBR	9.3	8.5	9.3	14.9	15.7	14.0	12.9	15.5	17.0	14.9	7.8	8.7	(36)
(37)			MCWBR	8.9	8.0	7.2	7.7	6.3	5.0	4.7	6.3	7.8	8.6	5.9	8.4	(37)
(38)		5% WB	MCDBR	9.4	8.4	8.2	14.0	12.8	11.5	10.7	13.4	14.7	13.9	7.4	8.3	(38)
(39)			MCWBR	9.1	8.0	6.5	7.8	6.1	5.0	4.9	5.9	7.2	8.4	5.7	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.333	0.384	0.408	0.410	0.410	0.434	0.416	0.373	0.355	0.336	0.304	(40)	
(41)		taud	2.020	1.958	1.910	2.137	2.194	2.279	2.210	2.234	2.302	2.238	2.076	1.974	(41)	
(42)		Ebn at Noon	627	726	768	817	837	841	810	803	793	712	573	523	(42)	
(43)		Edh at Noon	77	114	149	136	135	126	132	122	101	86	74	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.12	2.20	3.67	4.72	5.77	6.24	5.64	4.88	3.46	1.83	0.90	0.76	(44)	
(45)		RadStd	0.09	0.14	0.32	0.31	0.49	0.46	0.37	0.35	0.34	0.25	0.14	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.62	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page